

Result0 - 520 - gaussian_filter

Time34.12 msecond

Cycles34,609,734

Regs29

GPU0 - NVIDIA GeForce RTX 2060

SM Frequency1.01 cycle/nsecond

CC Process7.5 [B1327]project_canny

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GPU Speed of Light Throughput

High-level overview of the throughput for compute and memory resources of the GPU. For each unit, the throughput reports the achieved percentage of utilization with respect to the theoretical maximum. Breakdowns show the throughput for each individual sub-metric of Compute and Memory to clearly identify the highest contributor. High-level overview of the utilization for compute and memory resources of the GPU presented as a rooftop chart.

Compute (SM) Throughput [%]	64.66	Duration [msecond]	34.12
Memory Throughput [%]	64.66	Elapsed Cycles [cycle]	34,609,734
L1/TEX Cache Throughput [%]	97.53	SM Active Cycles [cycle]	34,434,217.70
L2 Cache Throughput [%]	0.63	SM Frequency [cycle/nsecond]	1.01
DRAM Throughput [%]	0.31	DRAM Frequency [cycle/nsecond]	5.49

Balanced Throughput

Compute and Memory are well-balanced. To reduce runtime, both computation and memory traffic must be reduced. Check both the [Compute Workload Analysis](#) and [Memory Workload Analysis](#) sections.

Roofline Analysis

The ratio of peak float (fp32) to double (fp64) performance on this device is 32:1. The kernel achieved 8% of this device's fp32 peak performance and 0% of its fp64 peak performance. See the [Kernel Profiling Guide](#) for more details on roofline analysis.

GPU Throughput

Compute (SM) [%]

Memory [%]

Speed of Light (SOL) [%]

Compute Throughput Breakdown

Memory Throughput Breakdown

Floating Point Operations Roofline

Performance [FLOP/s] (1e12)

Arithmetic Intensity [FLOP/byte]

Compute Workload Analysis

Detailed analysis of the compute resources of the streaming multiprocessors (SM), including the achieved instructions per clock (IPC) and the utilization of each available pipeline. Pipelines with very high utilization might limit the overall performance.

Executed ipc (Inst/cycle)	2.43	SM Busy [%]	60.82
Issued ipc Active [Inst/cycle]	2.43	Issue Slots Busy [%]	60.82

Balanced

ALU is the highest-utilized pipeline (53.8%) based on active cycles, taking into account the rates of its different instructions. It executes integer and logic operations. It is well-utilized, but should not be a bottleneck.

Pipe Utilization (% of active cycles)

ALU

FMA

FP64

Shared (FP16+Tensor)

Tensor (All)

Tensor (FP)

Tensor (NT)

Pipe Utilization (% of peak instructions executed)

LSU

ALU

FMA

XU

Uniform

ADU

CBU

GPU

FP64

TEX

Tensor (FP)

Tensor (NT)

Memory Workload Analysis

Detailed analysis of the memory resources of the GPU. Memory can become a limiting factor for the overall kernel performance when fully utilizing the involved hardware units (Mem Busy), exhausting the available communication bandwidth between those units (Mem Bandwidth), or by reaching the maximum throughput of issuing memory instructions (Mem Pipes Busy). Detailed tables with data for each memory unit.

Memory Throughput [Mbyte/second]	829.53	Mem Busy [%]	48.77
L1/TEX Hit Rate [%]	99.27	Max Bandwidth [%]	64.66
L2 Hit Rate [%]	94.20	Mem Pipes Busy [%]	64.66

L2 Load Access Pattern

The memory access pattern for loads from L1/TEX to L2 is not optimal. The granularity of an L1/TEX request to L2 is a 128-byte cache line. That is 4 consecutive 32-byte sectors per L2 request. However, this kernel only accesses an average of 1.2 sectors out of the possible 4 sectors per cache line. Check the [Source Locations](#) section for uncoalesced loads and try to minimize how many cache lines need to be accessed per memory request.

L2 Store Access Pattern

The memory access pattern for stores from L1/TEX to L2 is not optimal. The granularity of an L1/TEX request to L2 is a 128-byte cache line. That is 4 consecutive 32-byte sectors per L2 request. However, this kernel only accesses an average of 1.0 sectors out of the possible 4 sectors per cache line. Check the [Source Locations](#) section for uncoalesced stores and try to minimize how many cache lines need to be accessed per memory request.

Memory Chart

Kernel

Global

Local

Texture

Surface

Shared

L1/TEX Cache

L2 Cache

System Memory

Device Memory

Peer Memory

Shared Memory

Instructions

Requests

Wavefronts

% Peak

Bank Conflicts

L1/TEX Cache

Instructions

Requests

Wavefronts

% Peak

Sectors

Sectors/Req

Hit Rate

Bytes

Sector Misses to L2

% Peak to L2

Returns to SM

% Peak to SM

L2 Cache

Requests

Sectors

Sectors/Req

% Peak

Hit Rate

Bytes

Throughput

Sector Misses to Device

Sector Misses to System

Sector Misses to Peer

Device Memory

Load

% Peak

Bytes

Throughput

Scheduler Statistics

Summary of the activity of the schedulers issuing instructions. Each scheduler maintains a pool of warps that it can issue instructions for. The upper bound of warps in the pool (Theoretical Warps) is limited by the launch configuration. On every cycle each scheduler checks the state of the allocated warps in the pool (Active Warps). Active warps that are not stalled (Eligible Warps) are ready to issue their next instruction. From the set of eligible warps the scheduler selects a single warp from which to issue one or more instructions (Issued Warp). On cycles with no eligible warps, the issue slot is skipped and no instruction is issued. Having many skipped issue slots indicates the combined values.

Warp Cycles Per Issued Instruction [cycle]	12.57	Avg. Active Threads Per Warp	32
Warp Cycles Per Executed Instruction [cycle]	12.57	Avg. Not-Predicted Off-Threads Per Warp	31.42

lg.Shuttle

On average, each warp of this kernel spends 5.6 cycles being stalled waiting for the local/global instruction queue to be not full. This represents about 44.2% of the total average of 12.6 cycles between issuing two instructions. Typically this stall occurs only when executing local or global memory instructions extremely frequently. If applicable, consider combining multiple lower-width memory operations into fewer wider memory operations and try interleaving memory operations and math instructions.

Warp Stall

Check the [Source Locations](#) section for the top stall locations in your source based on sampling data. The [Kernel Profiling Guide](#) provides more details on each stall reason.

Warps Per Scheduler

GPU Maximum Warps Per Scheduler

Theoretical Warps Per Scheduler

Active Warps Per Scheduler

Eligible Warps Per Scheduler

Issued Warp Per Scheduler

Warp State Statistics

Analysis of the states in which all warps spent cycles during the kernel execution. The warp states describe a warp's readiness or inability to issue its next instruction. The warp cycles per instruction define the latency between two consecutive instructions. The higher the value, the more warp parallelism is required to hide this latency. For each warp, the chart shows the average number of cycles spent in that state per issued instruction. Stalls are not always impacting the overall performance nor are they completely avoidable. Only focus on stall reasons if the schedulers fail to issue every cycle. When executing a kernel with mixed library and user code, these metrics show the combined values.

Warp Cycles Per Issued Instruction [cycle]	12.57	Avg. Active Threads Per Warp	32
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Warp Stall

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Warp State (All Cycles)

Stall LG Throttle

Stall Not Selected

Stall Wait

Selected

Stall Long Scoreboard

Stall Math Pipe Throttle

Stall Dispatch Stall

Stall MID Throttle

Stall Short Scoreboard

Stall Branch Resolving

Stall No Instruction

Stall Drain

Stall IMC Miss

Stall Misc

Stall Barrier

Stall Member

Stall Sleeping

Stall Tex Throttle

Instruction Statistics

Statistics of the executed low-level assembly instructions (EASIS). The instruction mix provides insight into the types and frequency of the executed instructions. A narrow mix of instruction types implies a dependency on few instruction pipelines, while others remain unused. Using multiple pipelines allows hiding latencies and enables parallel execution. Note that 'Instructions/Opcode' and 'Executed Instructions' are measured differently and can diverge if cycles are spent in system calls.

Executed Instructions [Inst]	2,513,203,200	Avg. Executed Instructions Per Scheduler [Inst]	20,943,360
Issued Instructions [Inst]	2,513,206,438	Avg. Issued Instructions Per Scheduler [Inst]	20,943,366.98

Executed Instruction Mix

IMADD

ISETP

LDS

IADDS

LOP3

PRMT

PLDP3

IQF

FFMA

BRA

ULDC

SHF

S2R

UIADDS

MOV

EXIT

STGA

ULEA

IMMXX

F2I

NVLink Topology

NVLink Topology diagram shows logical NVLink connections with transmit/receive throughput.

NVLink Topology

The system does not have any NVLink connections.

NVLink Tables

Detailed tables with properties for each NVLink.

Logical NVLink Properties

The system does not have any NVLink connections.

Launch Statistics

Summary of the configuration used to launch the kernel. The launch configuration defines the size of the kernel grid, the division of the grid into blocks, and the GPU resources needed to execute the kernel. Choosing an efficient launch configuration maximizes device utilization.

Grid size	57,600	Function Cache Configuration	CachePreference
Registers Per Thread [register/thread]	29	Static Shared Memory Per Block [byte/block]	0
Block Size	256	Dynamic Shared Memory Per Block [byte/block]	0
Threads [thread]	14,745,600	Driver Shared Memory Per Block [byte/block]	0
Waves Per SM	480	Shared Memory Configuration Size [Kbyte]	32.77

Occupancy

Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicates highly imbalanced workloads.

Theoretical Occupancy [%]	100	Block Limit Registers [block]	8
Theoretical Active Warps per SM [warp]	32	Block Limit Shared Mem [block]	16
Achieved Occupancy [%]	95.58	Block Limit Warps [block]	4
Achieved Active Warps per SM [warp]	30.58	Block Limit SM [block]	16

Occupancy Limiters

This kernel's theoretical occupancy is not impacted by any block limit.

Impact of Varying Register Count Per Thread

Warp Occupancy

Registers Per Thread

Impact of Varying Block Size

Warp Occupancy

Block Size

Impact of Varying Shared Memory Usage Per Block

Warp Occupancy

Shared Memory Per Block

Source Counters

Source metrics, including branch efficiency and sampled warp stall reasons. Warp Stall Sampling metrics are periodically sampled over the kernel runtime. They indicate when warps were stalled and couldn't be scheduled. See the documentation for a description of all stall reasons. Only focus on stalls if the schedulers fail to issue every cycle.

Branch Instructions [Inst]	80,179,200	Branch Efficiency [%]	100
Branch Instructions Ratio [%]	0.03	Avg. Divergent Branches	0

Uncoalesced Global Accesses

This kernel has uncoalesced global accesses resulting in a total of 322856660 excessive sectors (49% of the total 655771860 sectors). Check the L2 Theoretical Sectors Global Excessive table for the primary source locations. The [CUDA Programming Guide](#) had additional information on reducing uncoalesced device memory accesses.

L2 Theoretical Sectors Global Excessive

Location

Value

Value (%)

Warp Stall Sampling (All Cycles)

Location

Value

Value (%)

Warp Stall Sampling (Not-issued Cycles)

Location

Value

Value (%)

Most Instructions Executed

Location

Value

Value (%)